

ACTIVITY SHEET ANSWERS

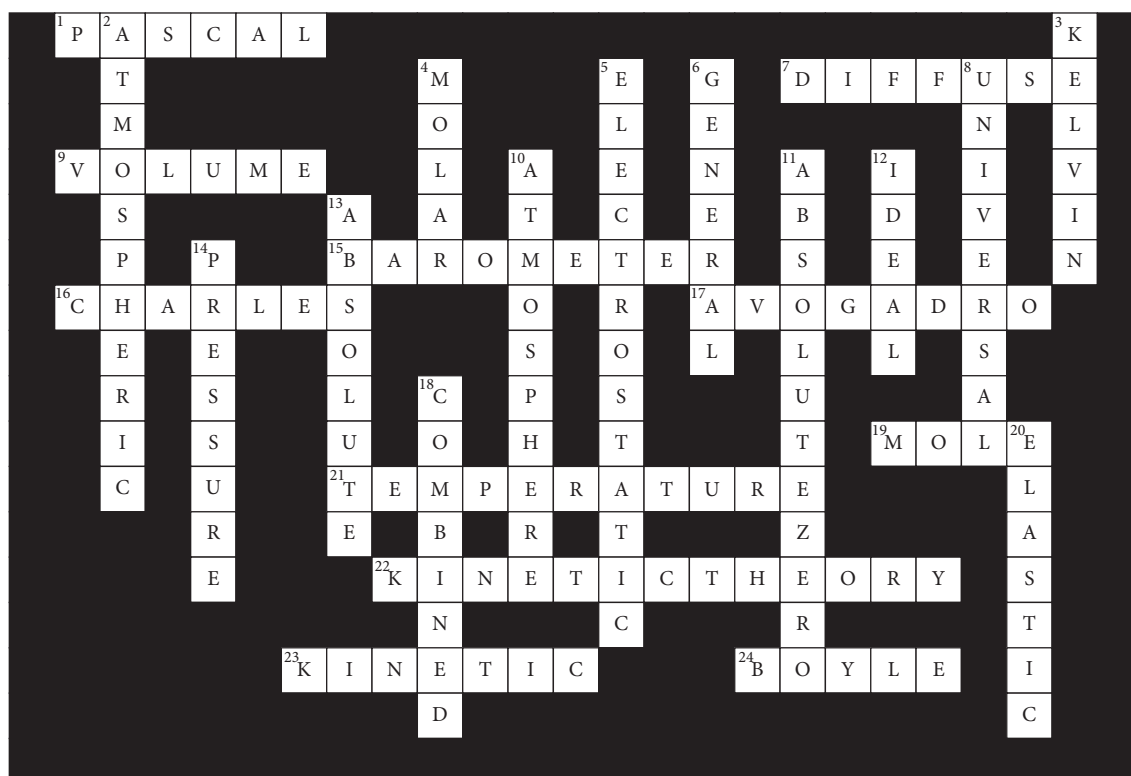
Chapter 8 Suggested answers

Prior knowledge

Beginning	What is all matter made of?	Particles / atoms	What are the three states of matter?	Solids, liquids and gases	What are the particles of a gas called?
Molecules	How do gas particles move?	At high speed in straight lines	How are gas particles spaced?	Well spaced out	What forces hold gas particles together?
None (forces between molecules are negligible)	What happens to a gas pumped into a container?	Gas fills container and takes its shape	What happens when gas particles collide?	Bounce off each other because of electrostatic repulsion	Do they lose energy when they collide?
No energy lost	What are these sorts of collisions called?	Elastic collisions	What do gases do to the surface of the container?	Create pressure	How do they do that?
By particles bouncing off container walls	How is pressure P defined?	Force exerted per unit area	What is the unit for pressure?	Pascal (Pa)	Why can gases be squashed?

Because particles are far apart and can be pushed closer	Are gases weightless?	No, they do have mass but are very light	What is the mixture of gases that we exist in called?	Atmosphere	What instrument is used to measure atmospheric pressure?
Barometer	What happens when you pump air into a tyre?	Pressure increases	What sort of energy do gas particles have?	Kinetic energy	Do gas particles have more or less energy than liquid particles?
More because they are moving much faster	How else are gas particles different?	They move independently of each other	What is the formula for kinetic energy?	$E_k = \frac{1}{2}mv^2$	What is temperature a measure of?
Average kinetic energy of particles of a substance	What does heating a gas do to the particles?	Makes them move faster	What does that do to their average kinetic energy?	Increases it rapidly	What happens as a result?
Temperature increases	What else happens to gases when they are heated?	They expand in volume if they can	Why?	Because the particles move faster and further apart	End

Activity sheet 8.1



Activity sheet 8.4

- 1 It stays pretty close to 1.
- 2 Nitrogen gas is behaving like an ideal gas under those conditions.
- 3 The compression factor rises steadily above 1 as the pressure increases above 200 bar.
- 4 Nitrogen gas is no longer behaving as an ideal gas. As the particles are forced closer, the volume occupied by the molecules themselves relative to the volume of the container becomes significant.
- 5 At 200 K, the compression factor drops significantly and then returns to 1 as the pressure increases up to 200 bar. At the even colder temperature of 100 K, the compression factor drops dramatically before rising up to 1 again.
- 6 Again, nitrogen gas is not behaving as an ideal gas. Intermolecular forces become significant as the particles are moving much more slower at lower temperatures, having much less kinetic energy.
- 7 At 400 K the compression factor remains close to 1 as the pressure rises up to 200 bar.
- 8 Above 200 bar the compression factor keeps rising above 1 for all temperatures.
- 9 At very high pressures, the particles are forced much closer together and the particles themselves occupy much of the available volume.
- 10 The steadily increasing deviation of the compression factor from 1 as the pressure increases above 200 bar becomes less marked as the compression factor is measured at higher temperatures.

Chapter 8 Revision

1 a $\rightarrow P$, b $\rightarrow L$, c $\rightarrow L$, d $\rightarrow E$, e $\rightarrow B$, f $\rightarrow C$, g $\rightarrow M$, h $\rightarrow O$, i $\rightarrow D$, j $\rightarrow N$, k $\rightarrow H$, l $\rightarrow K$, m $\rightarrow R$, n $\rightarrow F$, o $\rightarrow T$, p $\rightarrow A$, q $\rightarrow Q$, r $\rightarrow G$, s $\rightarrow J$, t $\rightarrow S$

2 a 0°C is equal to 273 K ; 0 K is equal to -273°C

b Boyle's law states that, at a constant temperature, volume is inversely proportional to applied pressure. Charles' law states that at a constant pressure, volume is directly proportional to the absolute temperature.

	Solid	Liquid	Gas
Spacing	Close together	Close together	Far apart
Arrangement	Regular pattern	Random arrangement	Random arrangement
Movement	Vibrate on the spot	Move around each other	Move rapidly and freely

Property	Assumptions about particles that explain property	
Compressible	far apart	
Diffuse rapidly	move rapidly	far apart
Mix readily	move rapidly	far apart
Low density	far apart	
Exert pressure	fast moving	elastic collisions
Fill container	move rapidly	no forces of attraction

5 a $0.75 \times 101.325 = 76\text{ kPa}$

b $0.75 \times 760 = 570\text{ mmHg}$

c $98 \div 101.325 = 0.97\text{ atm}$, $0.97 \times 760 = 740\text{ mmHg}$

6 a Because individual molecules have differing amounts of kinetic energy.

b $\sim 350\text{ m s}^{-1}$ for O_2 ; $\sim 400\text{ m s}^{-1}$ for N_2 ; $\sim 500\text{ m s}^{-1}$ for H_2O ; $\sim 1100\text{ m s}^{-1}$ for He ;
 $\sim 1650\text{ m s}^{-1}$ for H_2

c $M_r(\text{O}_2) = 2 \times 16.00 = 32.00$

$M_r(\text{N}_2) = 2 \times 14.01 = 28.02$

$M_r(\text{H}_2\text{O}) = 2 \times 1.01 + 16.00 = 18.02$

$A_r(\text{He}) = 4.00$

$M_r(\text{H}_2) = 2 \times 1.01 = 2.02$

d As the molecular/atomic mass of a gas decreases/increases, the average molecular speed increases/decreases.

e The heavier the particles, the less variation in speed occurs.

7 a 298 K

b -248°C

c 523 K

d -23°C

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Name	Statement	Formula	Alternatives	SI Units	Conditions
Boyle	Volume inversely proportional to pressure	$\frac{V \propto 1}{P}$	$P_1 V_1 = P_2 V_2$	V in L P in kPa	given mass constant temperature
Charles	Volume linearly proportional to temperature	$V \propto T$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	V in L T in K	given mass constant pressure
Combined	Relates pressure, volume and temperature	$PV \propto T$	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$	T in K V in L P in kPa	given mass
General	Relates pressure, volume, temperature and number of moles	$PV = nRT$	$P = nRT \div V$ $V = nRT \div P$ $T = PV \div nR$ $n = PV \div RT$	T in K V in L P in kPa n in mol $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$	

9 a $P_1 V_1 = P_2 V_2$

b $V_1 = 10.4 \text{ L}, P_1 = 90.2 \text{ kPa}, V_2 = ? \text{ L}, P_2 = 101.3 \text{ kPa}$

c Units for volumes and pressures are consistent.

d $V_2 = P_1 V_1 \div P_2 = 90.2 \times 10.4 \div 101.3 = 9.26 \text{ L}$

10 a $V_1 \div T_1 = V_2 \div T_2$

b $V_1 = 175 \text{ mL}, T_1 = 72^\circ\text{C}, V_2 = ? \text{ mL}, T_2 = 25^\circ\text{C}$

c $T_1 = 72 + 273 = 345 \text{ K}, T_2 = 25 + 273 = 298 \text{ K}$

d Units for volumes are consistent.

e $V_2 = T_2 V_1 \div T_1 = 298 \times 175 \div 345 = 151 \text{ mL}$

11 a $P_1 V_1 \div T_1 = P_2 V_2 \div T_2$

b $P_1 = 156 \text{ kPa}, V_1 = 25.6 \text{ mL}, T_1 = 45^\circ\text{C}, P_2 = 101.3 \text{ kPa}, V_2 = ? \text{ mL}, T_2 = 25^\circ\text{C}$

c $T_1 = 45 + 273 = 318 \text{ K}, T_2 = 25 + 273 = 298 \text{ K}$

d Units for volumes and pressures are consistent.

e $V_2 = T_2 P_1 V_1 \div P_2 T_1 = 298 \times 156 \times 25.6 \div 101.3 \times 298 = 36.9 \text{ mL}$

12 a $PV = nRT$

b $P = 101.3 \text{ kPa}, V = ? \text{ L}, T = 25^\circ\text{C}, n = 1.25 \text{ mol}, R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

c $T = 25 + 273 = 298 \text{ K}$

d Pressure in kPa, volume in L, temperature in K, n in mol

e $V = nRT \div P = 1.25 \times 8.31 \times 298 \div 101.3 = 30.6 \text{ L}$